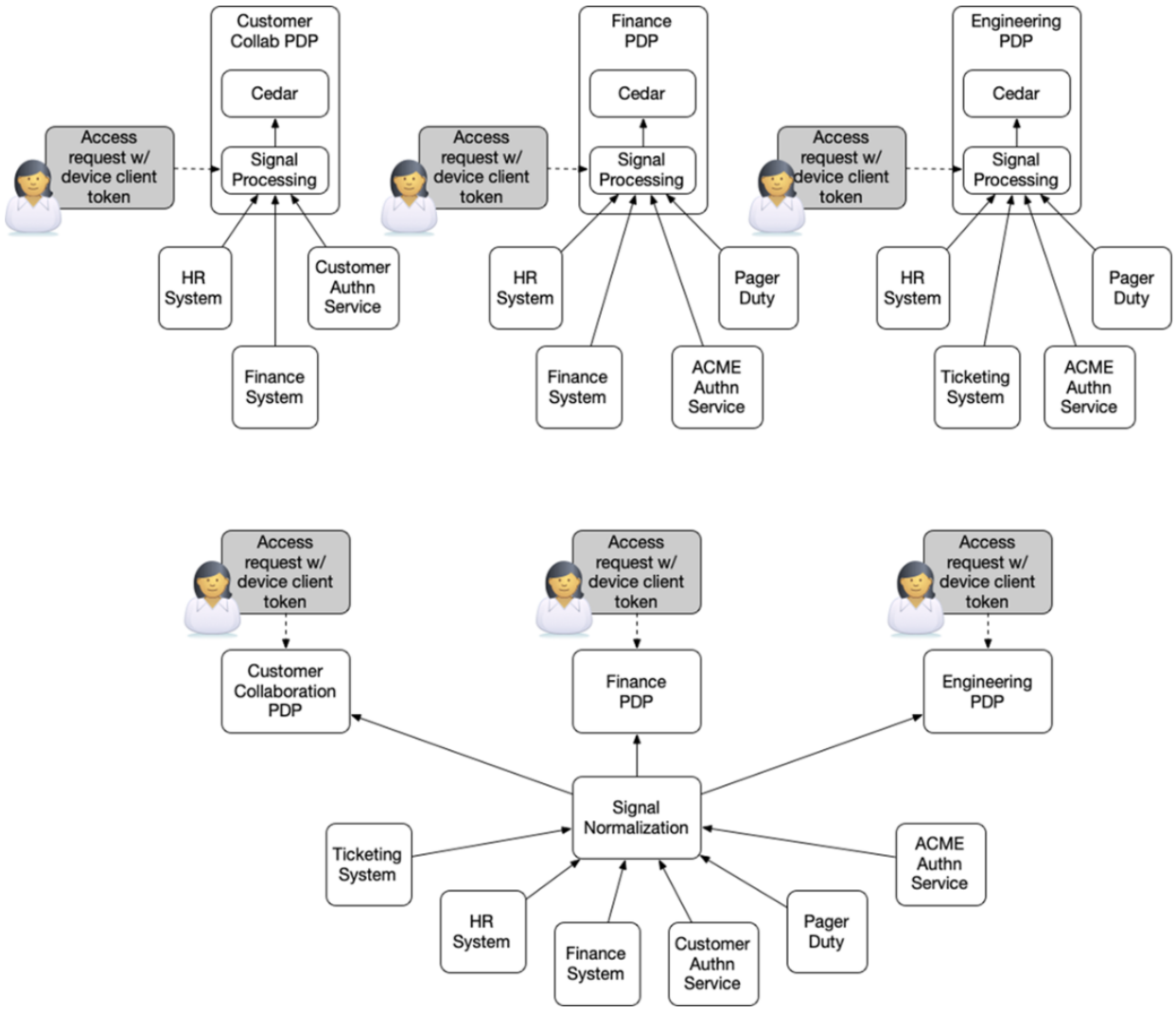


Figure 12.3 ACME's normalization architecture. Before normalization (top), each PDP integrated directly with HR, Finance, authentication, and operational systems (e.g., PagerDuty, ticketing), each with custom schemas and caches. After normalization (bottom), a shared signal-normalization layer aggregates and standardizes those feeds, delivering consistent context to all PDPs. Device posture arrives separately as a signed token on each access request and is verified alongside the normalized data. This hybrid model keeps tokens minimal (identifier + scopes) while preserving rich, consistent context for authorization decisions.

This architecture marked a turning point for ACME: signal normalization changed authorization from a collection of system-specific integrations into a unified, organization-wide service for providing trusted context.

12.6.2 Benefits of normalization

The impact was immediate. Policy authors could now reuse rules across systems because the underlying entity structure was consistent. Caching